

Troubleshooting: Unusable models

Daniel C. Reuman¹ and Angel Luis Robles Fernández²

¹Department of Ecology and Evolutionary Biology and Center for Ecological Research,
University of Kansas

²Department of Ecology and Evolutionary Biology and Center for Ecological Research,
University of Kansas

Abstract

When an xsdm model does not have a maximum likelihood, it should not be used. This document shows an example of how that can occur and makes some relevant comments.

The Eastern glass lizard, *Ophisaurus ventralis*, is found in the southeastern United States. We start by loading the data:

```
env_array = readRDS("ophisaurus_ventralis_env.Rds")
dim(env_array)

## [1] 2728    39     6

dimnames(env_array)[[3]]

## [1] "BIO01" "BIO10" "BIO11" "BIO12" "BIO16" "BIO17"

occ = readRDS("ophisaurus_ventralis_occ.Rds")
length(occ)

## [1] 2728
```

Here, there are 6 environmental variables recorded for 39 years in 2728 locations, with accompanying detections and pseudo-absences in the variable `occ`. The first three environmental variables (BIO1, BIO10, BIO11) are temperature variables, and the last three (BIO12, BIO16, BIO17) are precipitation variables. BIO1 is mean annual temperature, BIO10 is mean temperature of the warmest quarter, BIO11 is mean temperature of the coldest quarter, BIO12 is annual precipitation, BIO16 is precipitation of the wettest quarter, and BIO17 is precipitation of the driest quarter.

Now look at the distributions of values of environmental variables to make sure they are not on wildly different scales, which would cause problems for optimization:

```
apply(FUN=quantile, X=env_array, MARGIN=3, prob=c(.025, .25, .5, .75, .975))

##          BIO01      BIO10      BIO11      BIO12      BIO16      BIO17
## 2.5%  12.21381 21.39449  3.847143  88.08287 10.14201  2.213420
```

```
## 25%    16.87778 25.76851  8.884324 114.44185 14.34298  4.366665
## 50%    18.71020 26.63721 11.528053 131.57598 17.40133  5.776966
## 75%    20.42655 27.33272 14.381416 151.15700 20.91240  7.222351
## 97.5%  24.29674 28.29841 20.728660 195.81223 29.55371 10.721530
```

These distributions look basically OK.

Now fit 15 models, each from 25 starting conditions:

```
models = matrix(c(1,0,0,0,0,0,
                  0,1,0,0,0,0,
                  0,0,1,0,0,0,
                  0,0,0,1,0,0,
                  0,0,0,0,1,0,
                  0,0,0,0,0,1,
                  1,0,0,1,0,0,
                  1,0,0,0,1,0,
                  1,0,0,0,0,1,
                  0,1,0,1,0,0,
                  0,1,0,0,1,0,
                  0,1,0,0,0,1,
                  0,0,1,1,0,0,
                  0,0,1,0,1,0,
                  0,0,1,0,0,1), nrow=15, byrow=TRUE)
all_model_results = list()
for (i in 1:nrow(models))
{
  env_dat = env_array[, ,models[i,]==1,drop=FALSE]
  starts = xsdmMle::start_parms(env_dat,num_starts=25)
  all_optim_results = list()
  for (j in 1:nrow(starts))
  {
    all_optim_results[[j]] = optim(par=starts[j,],fn=xsdmMle::loglik_math,
                                   method="BFGS",
                                   env_dat=env_dat, occ=occ,negative=TRUE,
                                   control=list(trace=0))
  }
  all_model_results[[i]] = all_optim_results
}
```

Rank the models by BIC, bearing in mind that we've been working with the negative of the likelihood:

```
model_BICs = sapply(X=all_model_results,
                    FUN=function(x){
                      best_loglik = min(sapply(X=x, FUN=function(y){y$value}))
                      num_parms = length(x[[1]]$par)
                      n = length(occ)
```

```

        BIC = 2*best_loglik + num_parms*log(n)
        return(BIC)
    }
)

```

Also by AIC:

```

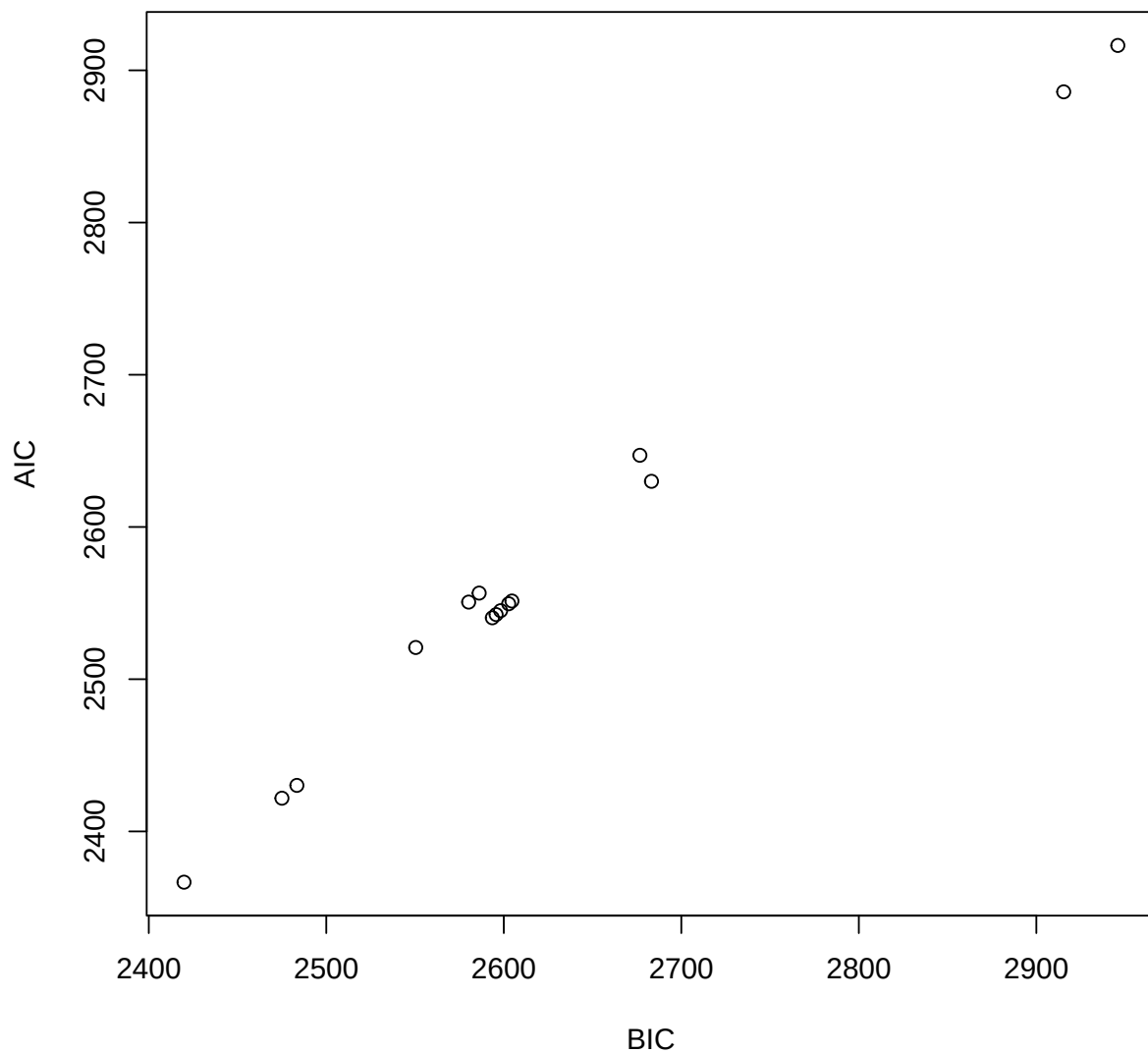
model_AICs = sapply(X=all_model_results,
  FUN=function(x){
    best_loglik = min(sapply(X=x, FUN=function(y){y$value}))
    num_parms = length(x[[1]]$par)
    AIC = 2*best_loglik + 2*num_parms
    return(AIC)
  }
)

inds = order(model_BICs)
rbind(model_BICs[inds],model_AICs[inds])

##           [,1]      [,2]      [,3]      [,4]      [,5]      [,6]      [,7]      [,8]
## [1,] 2419.854 2475.012 2483.436 2550.383 2580.205 2586.123 2593.541 2595.662
## [2,] 2366.653 2421.810 2430.234 2520.826 2550.648 2556.567 2540.339 2542.460
##           [,9]      [,10]      [,11]      [,12]      [,13]      [,14]      [,15]
## [1,] 2598.207 2602.755 2604.566 2676.652 2683.193 2915.417 2945.895
## [2,] 2545.005 2549.553 2551.364 2647.095 2629.991 2885.860 2916.339

plot(model_BICs,model_AICs,type="p",xlab="BIC",ylab="AIC")

```



```
order(model_BICs)
## [1] 11  8 14  5  3  1 13 15  7 10  9  2 12  4  6

order(model_AICs)
## [1] 11  8 14  5 13 15  7 10  3  9  1 12  2  4  6
```

The AIC and BIC results are pretty well aligned, and the four best models are the same:

```
models[order(model_BICs)[1:4],]
##      [,1] [,2] [,3] [,4] [,5] [,6]
```



```
## [1,]    0    1    0    0    1    0
## [2,]    1    0    0    0    1    0
## [3,]    0    0    1    0    1    0
## [4,]    0    0    0    0    1    0
```

These four models all use the same predictor, the fifth one, and then various possible second predictors.

Among the models we fitted, there is one clear winner in BIC, model 11. So let's try to use it. Start by optimizing it a bit harder to see if we can do any better.

```
i=11
env_dat = env_array[, ,models[i,]==1,drop=FALSE]
starts = xsdmMle::start_parms(env_dat,num_starts=100)
model_11_results = list()
for (j in 1:nrow(starts))
{
  model_11_results[[j]] = optim(par=starts[j,],fn=xsdmMle::loglik_math,
                                method="BFGS",
                                env_dat=env_dat, occ=occ,negative=TRUE,
                                control=list(trace=0))
}
min(sapply(X=all_model_results[[11]], FUN=function(y){y$value}))

## [1] 1174.326

min(sapply(X=model_11_results, FUN=function(y){y$value}))

## [1] 1174.34
```

Pretty similar, so move forward by looking at the results for this model:

```
examine_optim_results = function(optim_results,mask=NULL)
{
  #put optimization results in order from best to worst
  bestlogliks = sapply(X=optim_results,FUN=function(x){x$value})
  inds = order(bestlogliks)
  bestlogliks = bestlogliks[inds]
  optim_results = optim_results[inds]

  #model convergence
  convergences = sapply(X=optim_results,FUN=function(x){x$convergence})

  #compute distances to the best result in parameter space
  best_parms_math = optim_results[[1]]$par
  parms_dists_to_best = lapply(
    X=optim_results,
    FUN=function(x){
      xsdmMle::distance_between_params_hungarian(
```

```

    x$par,
    best_parms_math,
    mask=mask,
    GiveClosestRep=TRUE)
  }
)
parms_dists = sapply(X=parms_dists_to_best, FUN=function(x){x$distance})

#look at the best 5 results in parameter space
bestparms=sapply(X=parms_dists_to_best, FUN=function(x){unlist(x$representative)})

#put it all together
return(rbind(bestlogliks,convergences,parms_dists,bestparms))
}

h=examine_optim_results(all_model_results[[11]])
t(h[,1:8])

##      bestlogliks  convergences  parms_dists      mu1      mu2  sigltil1
## [1,]    1174.326            0    0.000000  29.73088  51.17655  0.3144583
## [2,]    1174.409            0    3.637603  29.47053  47.77001  0.3188141
## [3,]    1174.491            1    6.528463  29.29029  45.04127  0.3267773
## [4,]    1174.651            0   10.073297  29.02536  41.69584  0.3223081
## [5,]    1174.704            0   11.014513  28.95176  40.81217  0.3213432
## [6,]    1174.713            1   11.185832  28.94906  40.65592  0.3195780
## [7,]    1174.815            1   11.765075  28.86203  40.02645  0.3260034
## [8,]    1177.376            1   25.485343  27.96436  27.03693  0.2924514
##      sigltil2  sigrtil1 sigrtil2      ctil      pd      o_mat1      o_mat2
## [1,] 1.384285e+03 0.6457177 6.999370 -13.793627 0.5407866 -0.9965635 0.08283238
## [2,] 7.284659e+00 0.6405230 6.660255 -12.553030 0.5423771 -0.9964628 0.08403461
## [3,] 7.340614e+01 0.6320182 6.362509 -11.609792 0.5435305 -0.9962320 0.08672833
## [4,] 3.051941e+03 0.6411497 5.967987 -10.464529 0.5446594 -0.9962109 0.08697070
## [5,] 7.046120e+07 0.6443363 5.862402 -10.148548 0.5451424 -0.9962183 0.08688525
## [6,] 3.253548e+05 0.6477503 5.848594 -10.075503 0.5456345 -0.9962077 0.08700733
## [7,] 1.806372e+02 0.6383796 5.683158 -10.143547 0.5423364 -0.9962424 0.08660851
## [8,] 1.449955e+03 0.7118265 3.720807  -5.820755 0.5466379 -0.9954728 0.09504696
##      o_mat3      o_mat4
## [1,] -0.08283238 -0.9965635
## [2,] -0.08403461 -0.9964628
## [3,] -0.08672833 -0.9962320
## [4,] -0.08697070 -0.9962109
## [5,] -0.08688525 -0.9962183
## [6,] -0.08700733 -0.9962077
## [7,] -0.08660851 -0.9962424
## [8,] -0.09504696 -0.9954728

```

The very large values of `sigltil2` should be effectively equivalent to setting `sigltil2` to `Inf`. So we consider the corresponding boundary model:

```

i

## [1] 11

env_dat = env_array[,models[i,]==1,drop=FALSE]
mask=c(sigltil2=Inf)

new_starts = xsdmMle::start_parms(env_dat[occ==1,,drop=FALSE],
                                   mask=mask,num_starts=100)

new_starts

## # A tibble: 100 x 8
##      mu1    mu2 sigltil1 sigrttil1 sigrttil2    ctil    pd o_par1
##      <dbl> <dbl>    <dbl>    <dbl>    <dbl> <dbl> <dbl> <dbl>
##  1  26.0  19.3  -0.186    0.0387    1.34 -0.735 -0.446    3.68
##  2  27.0  25.8   0.507   -0.654    2.03 -1.26  1.75   -5.74
##  3  27.5  16.0   0.160   -0.308    2.38 -0.996  0.652    8.39
##  4  26.5  22.5  -0.533   -1.00    1.69 -1.52 -1.54   -1.03
##  5  26.7  14.4   0.334    0.212    2.21 -1.39 -0.996   -8.10
##  6  27.7  20.9  -0.359   -0.481    1.51 -0.865  1.20    1.33
##  7  27.2  17.6  -0.706   -0.135    1.17 -1.13  0.103   -3.39
##  8  26.2  24.2  -0.0128  -0.828    1.86 -0.605 -2.09    6.04
##  9  26.3  16.8  -0.619   -0.394    1.95 -0.670 -0.172   -4.57
## 10  27.4  23.3   0.0738   0.299    1.26 -1.19  2.03    4.86
## # i 90 more rows

bdry_optim_results = list()
for (j in 1:nrow(new_starts))
{
  #print(j)
  bdry_optim_results[[j]] = optim(par=new_starts[j,],fn=xsdmMle::loglik_math,
                                  method="BFGS",
                                  env_dat=env_dat,occ=occ,mask=mask,negative=TRUE,
                                  control=list(trace=0,maxit=500))
}

```

Have a look at these results:

```

h=examine_optim_results(bdry_optim_results,mask=mask)
t(h[,1:8])

##      bestlogliks convergences parms_dists    mu1    mu2 sigltil1 sigltil2
## [1,]    1174.392           0  0.0000000 29.55325 47.80564 0.3174066      Inf
## [2,]    1174.395           0  2.0219092 29.58171 49.54774 0.3233925      Inf
## [3,]    1174.432           0  0.8730595 29.48638 48.40804 0.3221770      Inf
## [4,]    1174.435           0  1.0380063 29.40231 46.81870 0.3223252      Inf
## [5,]    1174.445           0  0.4020091 29.44233 47.75971 0.3232318      Inf
## [6,]    1174.450           0  1.2584849 29.40669 46.56742 0.3245121      Inf

```

```
## [7,]      1174.459          0  1.7340485 29.34413 46.15780 0.3237365      Inf
## [8,]      1174.490          0  2.5653975 29.30161 45.37927 0.3185544      Inf
##      sigrttil1 sigrttil2      ctil      pd      o_mat1      o_mat2      o_mat3
## [1,] 0.6449675 6.691000 -12.47913 0.5434579 -0.9962686 0.08630729 -0.08630729
## [2,] 0.6265561 6.764974 -13.50229 0.5385020 -0.9964917 0.08369109 -0.08369109
## [3,] 0.6287720 6.643004 -13.10447 0.5388382 -0.9965038 0.08354714 -0.08354714
## [4,] 0.6391560 6.561144 -12.19965 0.5429631 -0.9964035 0.08473508 -0.08473508
## [5,] 0.6279646 6.579906 -12.85619 0.5389726 -0.9964684 0.08396858 -0.08396858
## [6,] 0.6282987 6.484128 -12.32867 0.5408820 -0.9962787 0.08619065 -0.08619065
## [7,] 0.6359417 6.486105 -11.98564 0.5433333 -0.9963895 0.08489943 -0.08489943
## [8,] 0.6461622 6.404934 -11.68508 0.5432430 -0.9963882 0.08491501 -0.08491501
##      o_mat4
## [1,] -0.9962686
## [2,] -0.9964917
## [3,] -0.9965038
## [4,] -0.9964035
## [5,] -0.9964684
## [6,] -0.9962787
## [7,] -0.9963895
## [8,] -0.9963882
```

The best likelihoods obtained for this boundary model are very similar to those obtained for the initial, non-boundary model, and the boundary model has one fewer parameter, so we tentatively adopt the boundary model over the earlier model. However, these optimizations results don't look great. Although the likelihoods of the top several optimization results are pretty similar, the corresponding parameters are rather more spread out in parameter space than one would expect if these distinct optimizations were all converging to the same clean likelihood peak. This is a sign that the likelihood surface may be ridged or flat around its highest values, and/or that we may not have successfully optimized.

To investigate the model further, we profile:

```
pnames=names(xsdmMle::make_mask_names(2))
pnames=pnames[pnames!="sigltil2"]

values = sapply(X=bdry_optim_results, FUN=function(x){x$value})
inds = order(values)
bdry_optim_results = bdry_optim_results[inds]

all_profiles=list()
linc = rep(0.05, length(pnames))
rinc = rep(0.05, length(pnames))
linc[8] = 0.001
rinc[8] = 0.001
for (counter in 1:length(pnames))
{
  all_profiles[[counter]] = xsdmMle::profile_likelihood(
    profile_parameter=pnames[counter],
```

```

        increment_left=linc[counter],
        increment_right=rinc[counter],
        num_steps_left=50,
        num_steps_right=50,
        alpha=0.95,
        optim_param_vector=bdry_optim_results[[1]]$par,
        env_dat=env_dat,
        occ=occ,
        mask=mask,
        num_threads=6
    )
}
names(all_profiles) = pnames

```

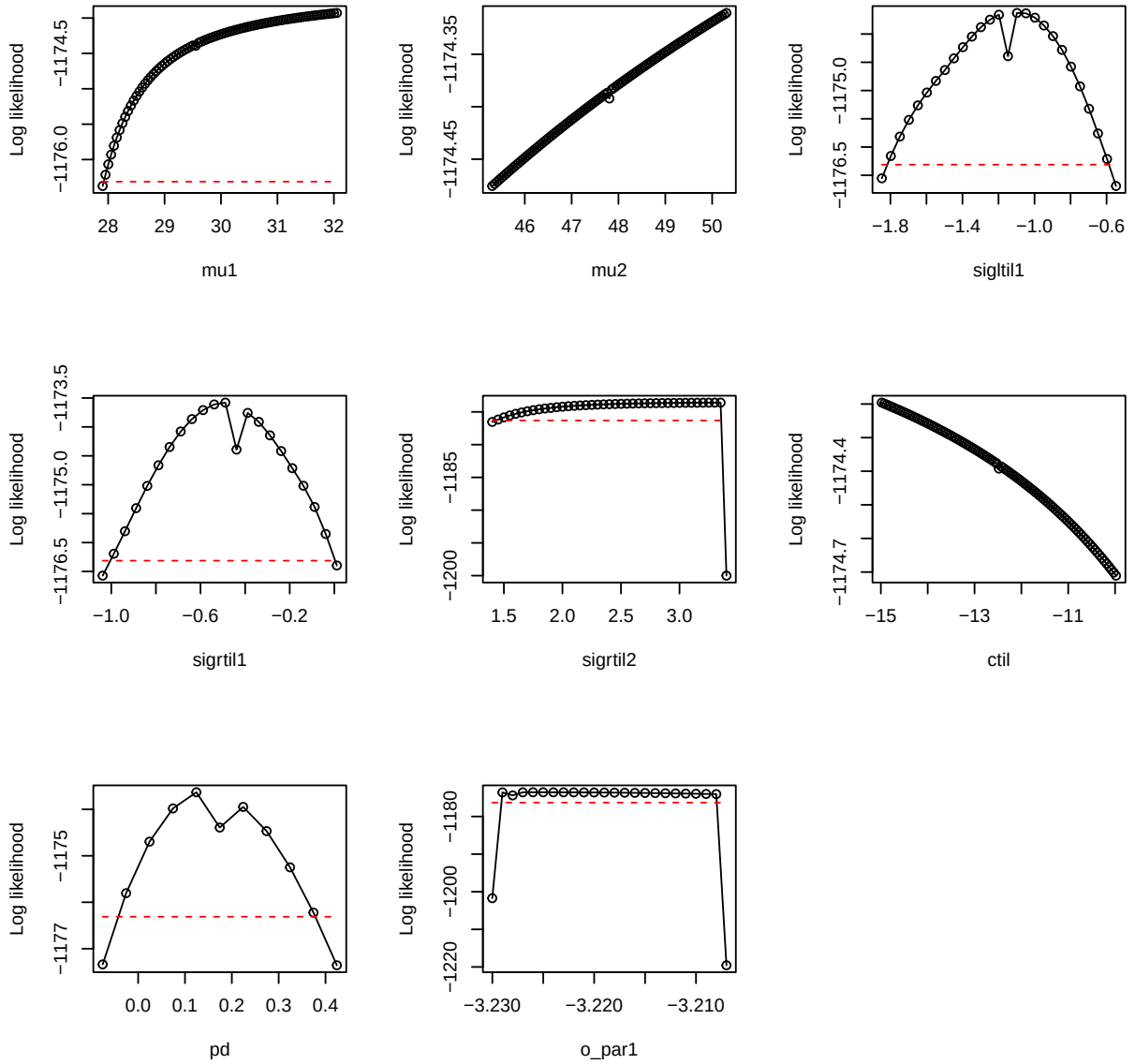
Now plot these profiles:

```

plot_tool=function(ap,index)
{
    x=ap[[index]]$profile$value_math
    y=ap[[index]]$profile$loglik
    xlab=names(ap)[index]
    thresh=ap[[index]]$threshold
    plot(x,y,
         type="o",xlab=xlab,
         ylab="Log likelihood")
    lines(range(x),rep(thresh,2),type="l",
         lty="dashed",col="red")
}

par(mfrow=c(3,3))
plot_tool(all_profiles,1)
plot_tool(all_profiles,2)
plot_tool(all_profiles,3)
plot_tool(all_profiles,4)
plot_tool(all_profiles,5)
plot_tool(all_profiles,6)
plot_tool(all_profiles,7)
plot_tool(all_profiles,8)

```



These profiles are not dome-shaped, and have other idiosyncratic features, confirming that we had not effectively optimized the likelihood. The μ_2 profile, in particular, shows problems.

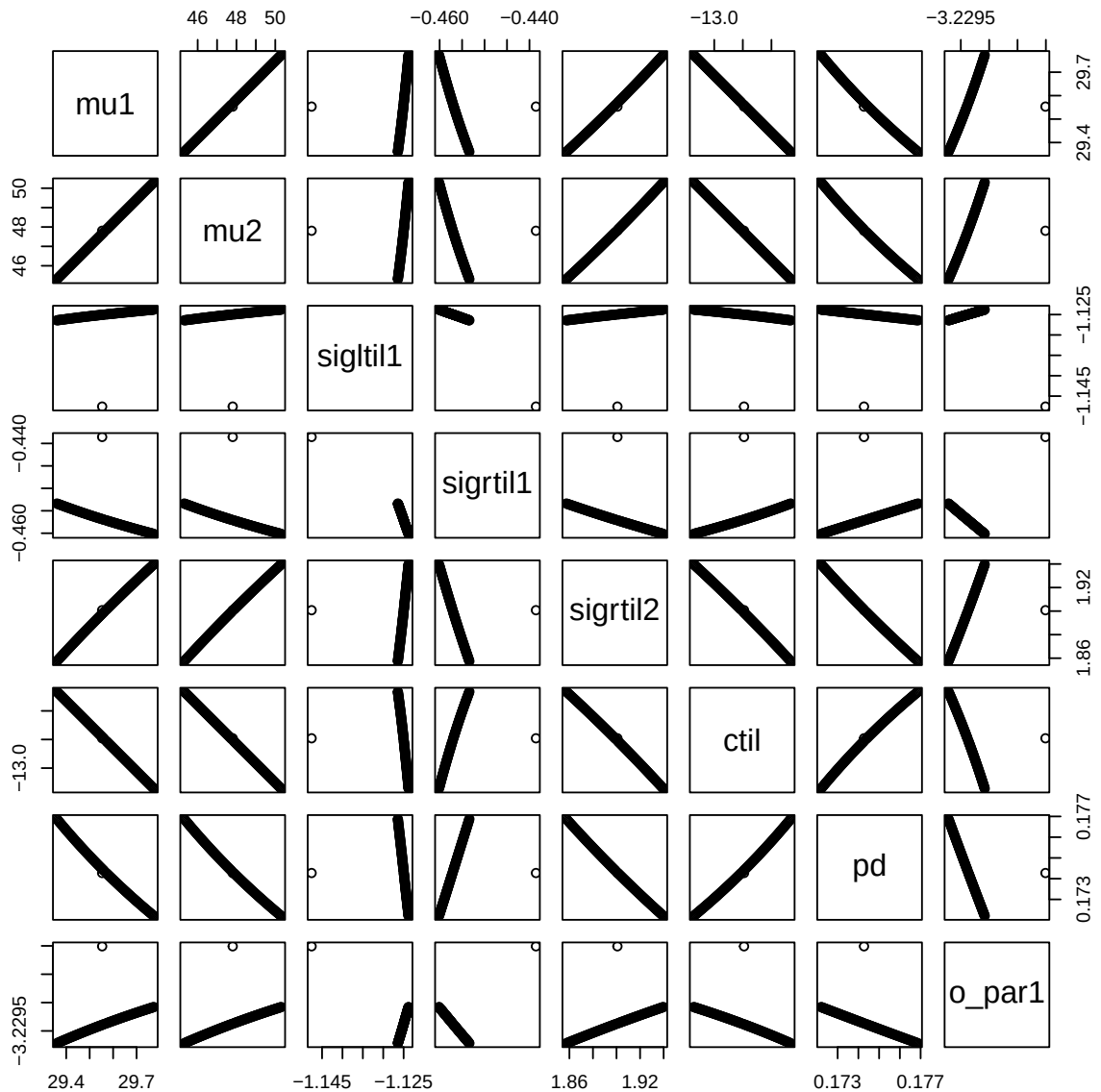
We do a pairs plot based on the μ_2 profile to investigate further:

```
p = all_profiles[[2]]$parameters
head(p)
```

##	mu1	mu2	sigltil1	sigrttil1	sigrttil2	ctil	pd	o_par1
## 1	29.35999	45.30564	-1.126416	-0.4533869	1.857404	-11.66308	0.1768780	-3.229717
## 2	29.36412	45.35564	-1.126384	-0.4534679	1.858298	-11.68000	0.1768217	-3.229710
## 3	29.36824	45.40564	-1.126352	-0.4535484	1.859190	-11.69693	0.1767656	-3.229703
## 4	29.37236	45.45564	-1.126321	-0.4536285	1.860081	-11.71386	0.1767098	-3.229695

```
## 5 29.37649 45.50564 -1.126289 -0.4537082 1.860971 -11.73079 0.1766543 -3.229688
## 6 29.38061 45.55564 -1.126259 -0.4537875 1.861859 -11.74772 0.1765989 -3.229680
## sigltil2
## 1 Inf
## 2 Inf
## 3 Inf
## 4 Inf
## 5 Inf
## 6 Inf
```

```
p = p[,1:8]
pairs(p)
```



This pairs plot helps us see the tradeoff going on between parameters. As μ_2 is increased, other parameters, especially `ctil`, change monotonically. These results strongly suggest a ridge in the likelihood surface that appears to rise asymptotically along some path in parameter space for which μ_2 is increasing. We have not identified a maximum of the likelihood function, and it looks as though there may not be one if the increase is indeed asymptotic.

Let's have a look at what the growth-environment function looks like for the best parameters we have found so far, maybe that will give some insight into what is going wrong.

```
param_list = bdry_optim_results[[1]]$par
param_list["sigltil2"] = Inf
param_list

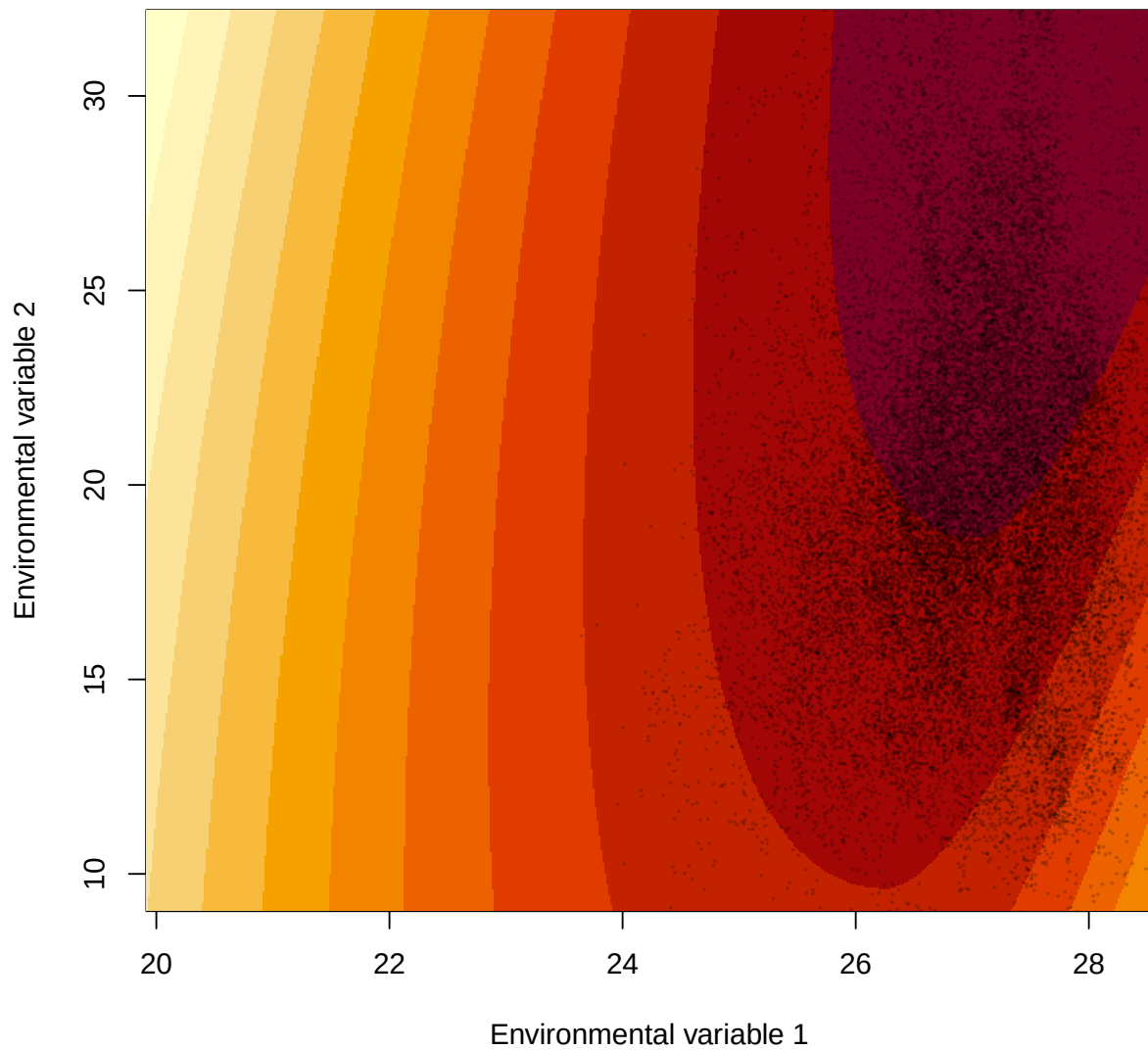
##          mu1          mu2    sigltil1    sigrtil1    sigrtil2          ctil
## 29.5532512 47.8056353 -1.1475716 -0.4385554  1.9007634 -12.4791250
##          pd          o_par1    sigltil2
##  0.1742715 -3.2280075          Inf

param_list_b = xsdmMle::math_to_bio(param_list)
param_list_b

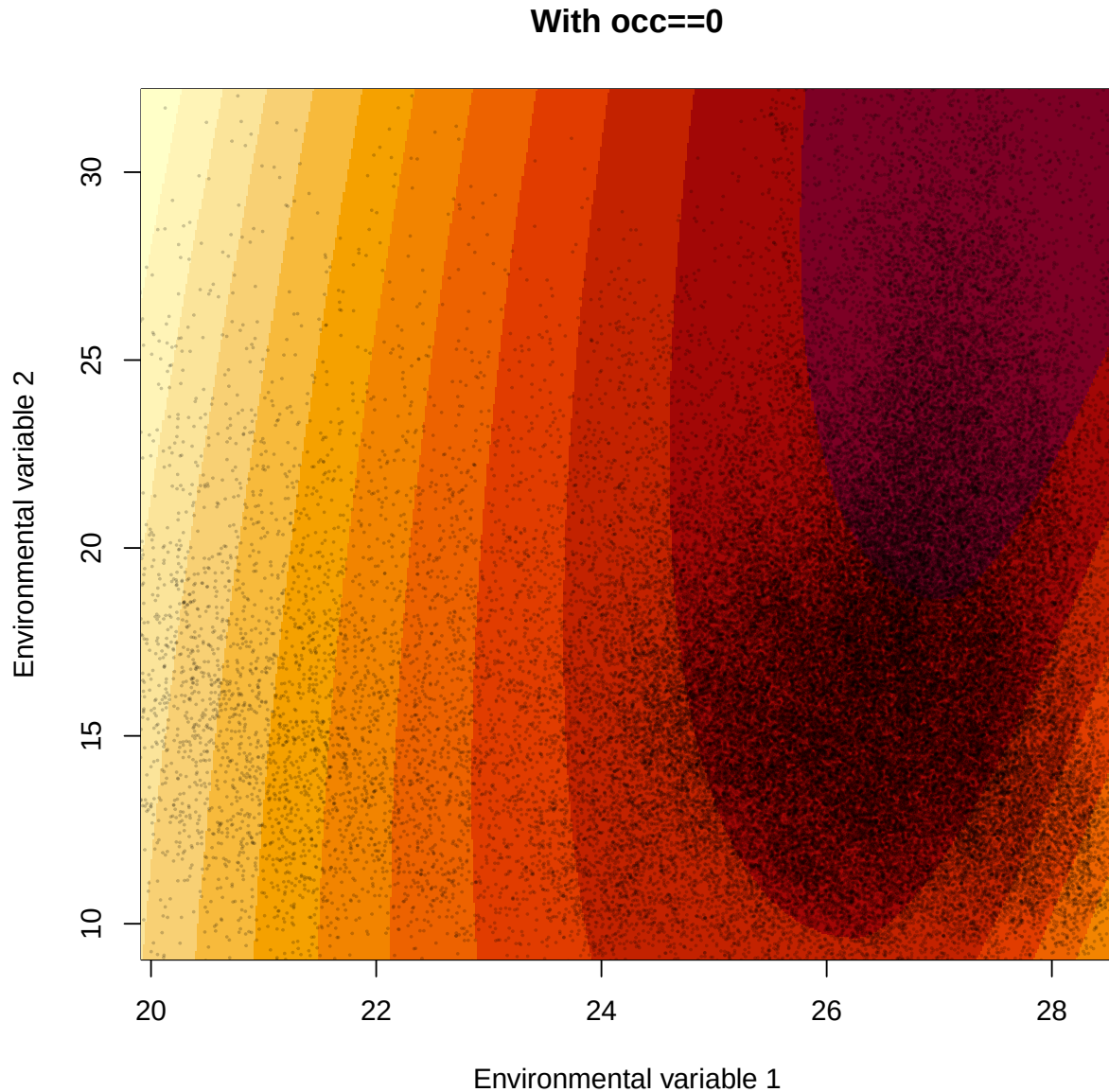
## $mu
## [1] 29.55325 47.80564
##
## $sigltil
## [1] 0.3174066          Inf
##
## $sigrtil
## [1] 0.6449675 6.6910004
##
## $ctil
## [1] -12.47913
##
## $pd
## [1] 0.5434579
##
## $o_mat
##          [,1]          [,2]
## [1,] -0.99626856 -0.08630729
## [2,]  0.08630729 -0.99626856

plot_lims = list(quantile(env_dat[,1]-param_list_b$mu[1],c(0.01,0.99)),
                  quantile(env_dat[,2]-param_list_b$mu[2],c(0.01,0.99)))
source("../R/interpret_parameters.R")
interpret_parameters(param_list=param_list_b,
                    plot_indices=c(1,2),
                    plot_lims=plot_lims,
                    main="With occ==1")
points(env_dat[occ==1,,1],env_dat[occ==1,,2],pch=20,
       col=rgb(0,0,0,.2),cex=0.2)
```


With occ==1



```
interpret_parameters(param_list=param_list_b,  
                    plot_indices=c(1,2),  
                    plot_lims=plot_lims,  
                    main="With occ==0")  
points(env_dat[occ==0,,1],env_dat[occ==0,,2],pch=20,  
       col=rgb(0,0,0,.2),cex=0.2)
```



Estimated values of μ_2 are outside the range of the environmental data. The actual distribution of the species is across Florida and along the coasts of the Southeastern United States, and the southern extent of the species is very likely constrained by the Atlantic Ocean and the Gulf of Mexico rather than by temperature and precipitation. Ultimately the modeling probably fails here for these reasons. A Bayesian approach to the same problem could likely address some of the limitations, here, by setting appropriate priors. In the frequentist setting, one must discard the model.

Immediate next steps should probably include examination of models 8 and 14, which are the next-best models according to BIC, to see if they have similar problems. For brevity and because we are just trying to illustrate statistical principles and workflows, we skip straight to examining model 14, which may be simpler because it only uses one environmental variable. The larger point is, that when a model does not have a maximum likelihood, it should be discarded in favor of others

models which do have maximum likelihoods. Examining model 5:

```
model_5_optim_results = all_model_results[[5]]
h=examine_optim_results(model_5_optim_results)
t(h[,1:8])
```

##		bestlogliks	convergences	parms_dists	mu	sigltil	sigrttil
##	[1,]	1255.413	0	0.0000000000	21.33107	2.399377	2.118516e+10
##	[2,]	1255.413	0	0.0004197823	21.33126	2.399545	2.920212e+06
##	[3,]	1255.413	0	0.0043086540	21.33485	2.400215	6.664093e+02
##	[4,]	1255.413	0	0.0026177745	21.33112	2.399409	3.823882e+02
##	[5,]	1255.413	0	0.0048557231	21.33091	2.399476	2.081598e+02
##	[6,]	1255.413	0	0.0050103145	21.33193	2.399567	2.028698e+02
##	[7,]	1255.413	0	0.0085723181	21.33750	2.400909	1.924301e+02
##	[8,]	1255.413	0	0.0077034714	21.32708	2.398714	1.601472e+02

##		ctil	pd	o_mat
##	[1,]	-1.806885	0.6389692	1
##	[2,]	-1.806516	0.6390226	1
##	[3,]	-1.808276	0.6388443	1
##	[4,]	-1.806788	0.6390008	1
##	[5,]	-1.806209	0.6390998	1
##	[6,]	-1.807133	0.6389728	1
##	[7,]	-1.809111	0.6387624	1
##	[8,]	-1.804798	0.6391885	1


```
values = sapply(X=model_5_optim_results, FUN=function(x){x$value})
inds = order(values)
model_5_optim_results = model_5_optim_results[inds]
```

One can see the boundary model with sigrttil to infinity should be considered:

```
i=5
env_dat = env_array[, ,models[i,]==1,drop=FALSE]
mask=c(sigrttil=Inf)
new_starts = xsdmMle::start_parms(env_dat[occ==1, , ,drop=FALSE],mask=mask,num_starts=100)

bdry_optim_results5 = list()
for (j in 1:nrow(new_starts))
{
  bdry_optim_results5[[j]] = optim(par=new_starts[j,],fn=xsdmMle::loglik_math,
                                  method="BFGS",
                                  env_dat=env_dat,occ=occ,mask=mask,negative=TRUE,
                                  control=list(trace=0,maxit=500))
}
```

Examine these results:

```

h=examine_optim_results(bdry_optim_results5,mask=mask)
t(h[,1:8])

##      bestlogliks convergences  parms_dists      mu sigltil sigrttil      ctil
## [1,]      1255.413           0 0.000000e+00 21.33077 2.399367      Inf -1.806580
## [2,]      1255.413           0 5.544855e-06 21.33077 2.399364      Inf -1.806580
## [3,]      1255.413           0 1.106750e-04 21.33084 2.399365      Inf -1.806666
## [4,]      1255.413           0 8.055958e-05 21.33085 2.399382      Inf -1.806604
## [5,]      1255.413           0 8.186739e-05 21.33085 2.399383      Inf -1.806602
## [6,]      1255.413           0 9.230715e-05 21.33086 2.399392      Inf -1.806604
## [7,]      1255.413           0 6.184925e-05 21.33078 2.399388      Inf -1.806519
## [8,]      1255.413           0 1.275815e-04 21.33086 2.399361      Inf -1.806675
##              pd o_mat
## [1,] 0.6390007      1
## [2,] 0.6390003      1
## [3,] 0.6389913      1
## [4,] 0.6390001      1
## [5,] 0.6389986      1
## [6,] 0.6390002      1
## [7,] 0.6390074      1
## [8,] 0.6389931      1

values = sapply(X=bdry_optim_results5, FUN=function(x){x$value})
inds = order(values)
bdry_optim_results5 = bdry_optim_results5[inds]

```

This model looks like it might have optimized well. Do profiles:

```

i=5
env_dat = env_array[, ,models[i,]==1,drop=FALSE]

all_profiles=list()
pnames=names(xsdmMle::make_mask_names(1))
pnames = pnames[pnames!="sigrttil"]
bdry_optim_results5[[1]]$par

##      mu1      sigltil1      pd      ctil
## 21.3307727 0.8752051 0.5710295 -1.8065797

linc = c(0.05,0.025,0.05,0.05)
rinc = c(0.15,0.025,0.05,0.05)
for (counter in 1:length(pnames))
{
  all_profiles[[counter]] = xsdmMle::profile_likelihood(
    profile_parameter=pnames[counter],
    increment_left=linc[counter],
    increment_right=rinc[counter],
    num_steps_left=50,

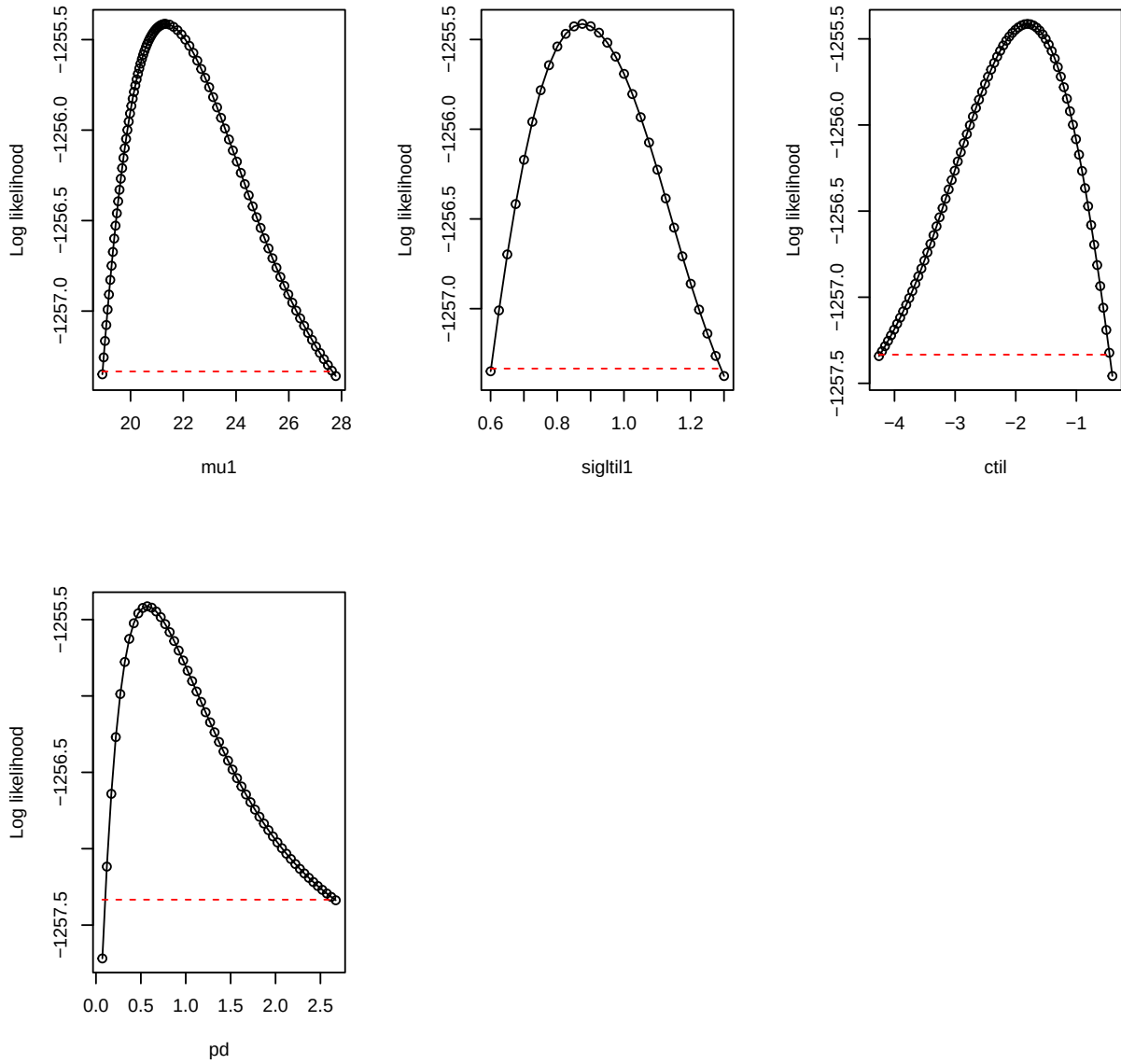
```

```

                                num_steps_right=50,
                                alpha=0.95,
                                optim_param_vector=bdry_optim_results5[[1]]$par,
                                env_dat=env_dat,
                                occ=occ,
                                mask=mask,
                                num_threads=6
                                )
}
names(all_profiles) = pnames

par(mfrow=c(2,3))
plot_tool(all_profiles,1)
plot_tool(all_profiles,2)
plot_tool(all_profiles,3)
plot_tool(all_profiles,4)

```



The profiles look adequate. So the model is suitable for use, unlike the previous models. If model 8 or 14 is suitable in the same manner, the lowest-BIC model suitable model should tend to be preferred.

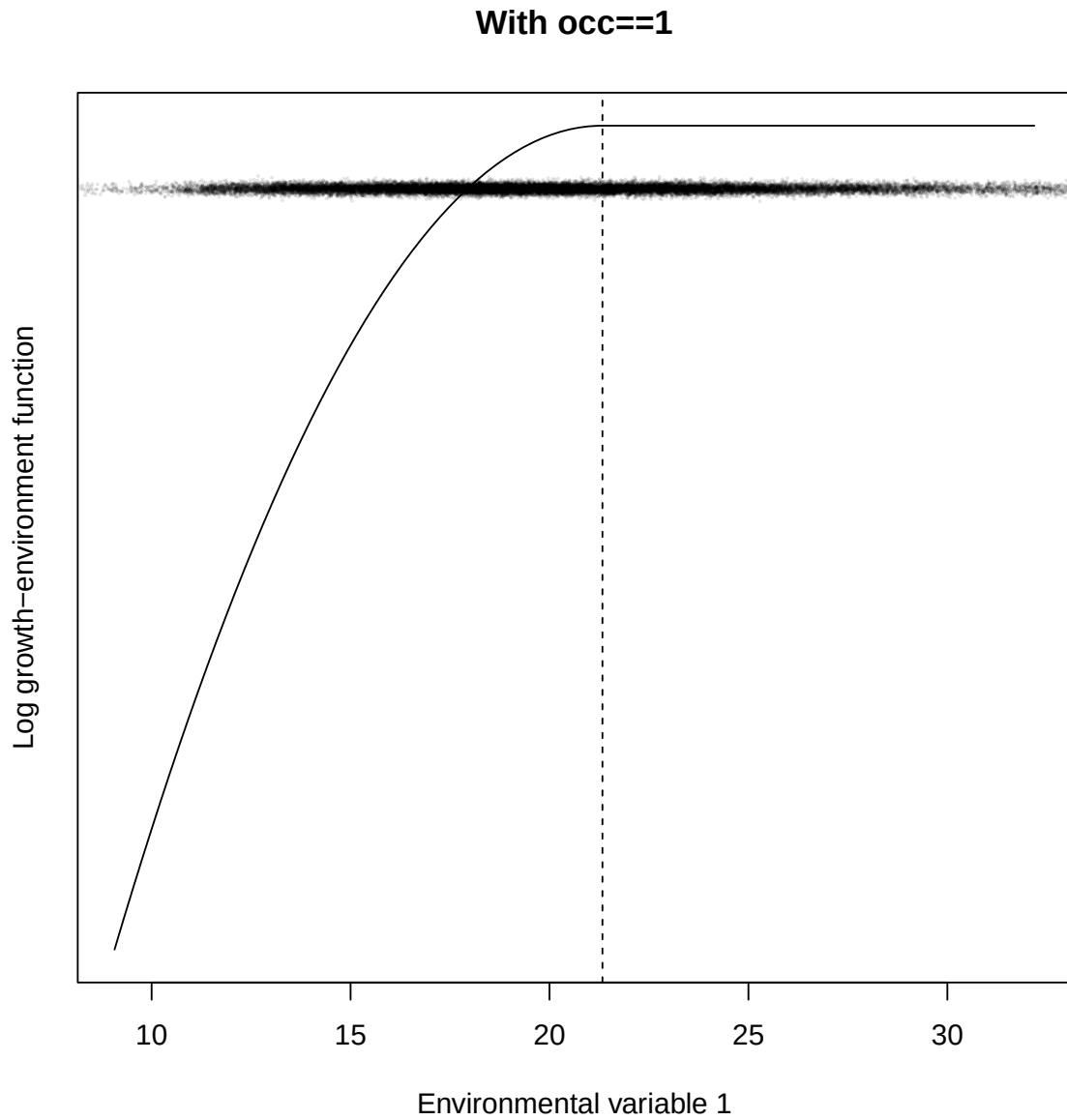
Examining the growth-environment function for this model:

```
param_list = bdry_optim_results5[[1]]$par
param_list_b = xsdmMle::math_to_bio(param_list)
param_list_b$sigrttil = Inf
plot_lims = list(quantile(env_dat[,1]-param_list_b$mu[1],c(0.01,0.99)))
interpret_parameters(param_list=param_list_b,
                     plot_indices=c(1),
                     plot_lims=plot_lims,
```

```

        main="With occ==1")
xvals = env_dat[occ==1,,1]
points(xvals,-2+rnorm(length(xvals),0,.1),pch=20,
       col=rgb(0,0,0,.1),cex=0.2)

```



```

interpret_parameters(param_list=param_list_b,
                    plot_indices=c(1),
                    plot_lims=plot_lims,
                    main="With occ==0")
xvals = env_dat[occ==0,,1]
points(xvals,-2+rnorm(length(xvals),0,.1),pch=20,
       col=rgb(0,0,0,.1),cex=0.2)

```

With occ==0

